

SIMATS ENGINEERING

ASSESSMENT TOOL 1 – Medium (Scenario Based Assessment)

COURSE CODE: CSA09

COURSE NAME: Programming in Java

COURSE OUTCOME COVERED

CO2: *Implement Collection Framework interfaces such as List, ArrayList, Set, Map, HashTable, and HashMap using iterators and generics to store, retrieve, and process groups of objects in web applications.(BL3,BL4)*

QUESTIONS: 5

Total Marks: 100

ANNEXURE A

Scenario Based Assessment

Code of Conduct:

I Sameer Ahmed G (192421154) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Sameer

RUBRICS FOR CALCULATION:

Criteria	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Context Analysis				
Application of Concepts				
Analytical & Decision-Making Skills				
Solution Design & Approach				
Clarity, Justification & Presentation				

Q1. Student Registration System

(a) List, Set, Map

(b) HashSet<Student> → Unique Students

HashMap<Student, ArrayList<String>> → Student → Courses

(c)

```
import java.util.*;

public class Main{

    public static void main(String[] args){

        HashSet<String> students=new HashSet<>();

        HashMap<String,ArrayList<String>> map=new HashMap<>();

        students.add("Sameer");

        students.add("Aman");

        map.put("Sameer",new ArrayList<>(Arrays.asList("Java","DBMS")));

        map.put("Aman",new ArrayList<>(Arrays.asList("Python","OS")));

        Iterator<String> it=students.iterator();

        while(it.hasNext()){

            String s=it.next();

            System.out.println(s+" -> "+map.get(s));

        }

    }

}
```

Output

Clear

```
Rahul -> [C++, CN]
Aman -> [Python, OS]
Sameer -> [Java, DBMS]

=== Code Execution Successful ===
```

Q2. E-Commerce Cart System

(a) ArrayList, LinkedList, Vector

(b) ArrayList → Product1 → Product2 → Product3

(c)

```
import java.util.*;

public class Main{

    public static void main(String[] args){

        ArrayList<String> cart=new ArrayList<>();

        cart.add("Laptop");

        cart.add("Mouse");

        cart.add("Keyboard");

        cart.remove("Mouse");

        Iterator<String> it=cart.iterator();

        while(it.hasNext())

            System.out.println(it.next());

    }

}
```

Output

Clear

```
Enter number of items: 3
Enter Item: Brush
Enter Item: Can
Enter Item: Hat
Enter item to remove: -
```

```
Cart Items:
Brush
Can
Hat
```

```
=== Code Execution Successful ===
```

Q3. Library Management System

(a) HashMap, TreeMap, Hashtable

(b) HashMap<ISBN,BookName>

(c)

```
import java.util.*;

public class Main{

    public static void main(String[] args){

        HashMap<String,String> books=new HashMap<>();

        books.put("101","Java");

        books.put("102","Python");

        System.out.println(books.get("101"));

        for(Map.Entry<String,String> b:books.entrySet())

            System.out.println(b.getKey()+" "+b.getValue());

    }

}
```

Output Clear

```
Enter Number of Books: 2
Enter ISBN: 1234
Enter Book Name: sameer
Enter ISBN: 1234567
Enter Book Name: Heights

Enter ISBN to Search: 1234
Book: sameer

All Books:
1234 : sameer
1234567 : Heights

=== Code Execution Successful ===
```

Q4. Employee Data Processing

(a) Type Safety, Reusability, Compile-time Error Checking

(b) ArrayList<Employee>

(c)

```
import java.util.*;

class Employee{
    String name; int salary;
    Employee(String n,int s){name=n;salary=s;}
}

public class Main{
    public static void main(String[] args){
        ArrayList<Employee> list=new ArrayList<>();
        list.add(new Employee("Sameer",60000));
        list.add(new Employee("Aman",40000));
        Iterator<Employee> it=list.iterator();
        while(it.hasNext()){
            Employee e=it.next();
            if(e.salary>50000)
                System.out.println(e.name+" "+e.salary);
        }
    }
}
```

Output

Clear

```
Enter Number of Employees: 2
Enter Name: sameer
Enter Salary: 12345
Enter Name: King
Enter Salary: 23456

Employees with Salary > 50000

=== Code Execution Successful ===
```

Q5. Online Voting System

(a) HashSet, HashMap, ArrayList

(b) HashSet → Voters

HashMap → Candidate → Votes

(c)

```
import java.util.*;

public class Main{

    public static void main(String[] args){

        HashSet<String> voters=new HashSet<>();

        HashMap<String,Integer> votes=new HashMap<>();

        String voter="Sameer";

        String candidate="A";

        if(voters.add(voter))

            votes.put(candidate,votes.getOrDefault(candidate,0)+1);

        else

            System.out.println("Already Voted");

        System.out.println(votes);

    }

}
```

Output

Clear

```
Enter Number of Voters: 2
Enter Voter Name: sameer
Enter Candidate: king
Enter Voter Name: sam
Enter Candidate: boom
```

Voting Results:

```
king : 1
boom : 1
```

=== Code Execution Successful ===